

Chris Dietrich

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<https://github.com/ChrisD790> | Website Portfolio: <https://chrisd790.github.io/Chris-Website/>

TECHNICAL SKILLS

Languages: C++, Python, C# , Java, SQL

Tools: GitHub, AutoCAD, SolidWorks, Xcelgo Experior

Technologies/Frameworks: Unity, OneDrive, Excel, REST API, ROS (learning), OpenCV (learning)

EDUCATION

Candidate for Bachelor of Applied Science University of Waterloo **2025 – Present**
Bachelor's of Applied Science in Mechatronics Engineering.

AFNORTH International Highschool Ontario Diploma (98.7% graduating average) **2021 – 2025**

WORK EXPERIENCE

Software Engineer Brock Solutions **Jan-Apr 2026**

- Built a C# tool enabling remote interaction with Experior (Xcelgo) airport baggage handling system simulations and emulations, allowing multiple engineers to access shared models and reducing licensing costs.
- Implemented REST API messaging between C# WinForms frontend and Experior C# backend.
- Developed a data synthesizing application using Microsoft SQL Client to query airport operational databases, call stored procedures, and synthesize real-time data into XML/CSV for simulation inputs, reducing data preparation time from 6 hours to 5 minutes.
- Created an AI-assisted C# GUI development workflow, doubling frontend productivity.
- Set up and networked Virtual Machines to implement airport baggage tracking software.

PROJECTS

Goose Hunt Eng-Hacks project **2026**
Developer

- Built a real-time multiplayer campus game using Python (FastAPI) and HTML/CSS/JavaScript, enabling cross-device gameplay over a shared network
- Implemented live location tracking and map visualization using the HTML5 Geolocation API and Leaflet
- Developed core game mechanics including dynamic shrinking zones, role management, and win condition logic
- Designed an interactive frontend and backend system to handle player state synchronization and real-time game flow
- Collaborated effectively using Git version control and team coordination to avoid merge conflicts

Unity game development **2024 – 2025**
Developer

- Developed two games in unity placing first at the Future Business

	- Leaders of America provincial competition - Utilized a Unity git repository for active collaboration with a Co-developer	2025 – Present
Voice assistant Developer	- Developed working Speech To Text (STT) and Text To Speech (TTS) system and custom wake word. - Developing a working voice assistant in Python to do arithmetic, simple computer functions and search queries. - Currently coding intent recognition on incoming STT queries.	
Autonomous Tic Tac Toe Robot Class Project	- Developed a fully autonomous robot to play Tic Tac Toe against a human player, using VEX IQ as the control hub, for the MTE100 class. - Used a shared GIT repository to keep our team's code in synch.	2025 – Present

EXTRACURRICULARS

Future Business Leaders of America (FBLA) club Club President	Led 30 members in fundraising efforts, collecting over \$2500 for student initiatives	2023 – 2025
FIRST Robotics Team Team Lead	- Led a team of six in developing a robot to compete in the FIRST Tech Challenge. - Developed several core components for the robots such as a grappling hook, angled slide, and claw designs winning our team the “Think” and “Innovate” awards. - Developed Java code for autonomous robot navigation, increasing average autonomous phase score by 20 points per match. - Won FIRST Tech scholarship valued at \$15000.	2022 – 2025
Waterloo Aerial Robotics Group (WARG) Member	Currently enrolled member of WARG Autonomy sub team planning to specialize in computer vision this term	2025 – Present

AWARDS

Schulich Leaders Scholarship	Selected as school nominee, for leadership and creativity.	2025
Valedictorian	Honored for academic excellence (98.7% average) and leadership.	2025
Graf Von Kielmannsegg	Recognized by <i>NATO Base commander General Guglielmo Luigi Miglietta</i> for academic excellence, community contribution, leadership and character.	2025
Future Business Leaders of America	First place awards for video game development using Unity X3, marketing, economics, coding and programming, international business, financial math, business management, and business education.	2023 - 2025